Transscleral Resection versus Iodine Brachytherapy for Choroidal Malignant Melanomas 6 Millimeters or More in Thickness

A Matched Case–Control Study

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Objective: To assess differences in treatment outcome after transscleral resection (TSR) and iodine brachytherapy (IBT) of choroidal melanomas too thick for ruthenium brachytherapy.

Design: Matched case–control study.

Setting: Two national ocular oncology services.

Participants: Forty-nine pairs of patients with choroidal melanoma 6 mm thick or more (range = 6–14), treated within 3 years of each other by either TSR or IBT, and retrospectively matched by age, visual acuity at diagnosis, largest basal tumor diameter, and posterior extension of the tumor.

Intervention: Iodine brachytherapy in Finland or TSR in United Kingdom. Prospectively collected time-to-event data were compared with parametric Weibull regression, adjusting for matching.

Main Outcome Measures: Tumor control, complications, and visual acuity.

Results: Risk of local recurrence after IBT was smaller than that after TSR (hazard ratio [HR] = 0.02, 95% confidence interval [CI] = 0.01–0.11, $P<0.001$, Weibull regression adjusted for matching), but the 8-year all-cause and melanoma-specific (HR = 0.81, 95% CI = 0.30–2.22, $P = 0.69$) survivals did not differ. The risks of cataract (HR = 2.05, 95% CI = 1.08–3.89, $P = 0.029$), maculopathy (HR = 2.28, 95% CI = 0.96–5.43, $P = 0.062$), and vitreous hemorrhage (HR = 2.30, 95% CI = 0.95–5.57, $P = 0.064$) were higher after IBT. Rubeosis, neovascular glaucoma, and optic neuropathy developed only after IBT. Risk of retinal detachment, exudative after IBT and rhegmatogenous after TSR (HR = 0.84, 95% CI = 0.40–1.75, $P = 0.63$), and risk of losing 20/60 vision (HR = 1.37, 95% CI 0.76–2.45, $P = 0.29$) were comparable between the groups, but risk of losing 20/200 vision was higher after IBT (HR = 2.38, 95% CI = 1.48–3.83, $P<0.001$). No overall difference in quality of life was found.

Conclusions: This study suggests that TSR preserves 20/200 vision better than IBT and avoids some of its major complications, but increases the risk of local recurrence. About 350 patients would need to be randomized to prove that TRS preserves 20/60 vision better than IBT. *Ophthalmology 2003;110:2235–2244 © 2003 by the American Academy of Ophthalmology.*

There is no consensus on the best treatment of choroidal melanomas with a thickness of 6 mm or more, except that ruthenium plaque brachytherapy (RuBT) is generally considered inadequate.[1–3] The therapeutic options include iodine[4–9] and palladium plaque brachytherapy,[10] charged particle teletherapy, stereotactic radiotherapy, transscleral local resection,[11,12] and enucleation. The published literature comparing any 2 of these treatments is scanty,[5,9,13–17] and thus there is a need for further controlled studies.

The best method of comparing 2 forms of treatment for

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choroidal melanoma would be a randomized prospective study.[5,9,16] Currently, this approach is seldom practical; however, because few centers offer more than one kind of conservative treatment for tumors of a given size, and those that do are unable to recruit sufficient numbers of patients within a reasonable time. One way to overcome this limitation—particularly when exploring which options for performing randomized prospective trials would likely be clinically most meaningful and practically most feasible—would be to perform multicenter case–control pilot studies.[14]

We used this approach to compare iodine 125 brachytherapy (IBT) with transscleral local resection (TSR). The former is currently the most widely used form of radiotherapy for thick uveal melanomas,[4–9] whereas the latter is an increasingly common, evolving technique that offers a possibility to avoid complications often associated with large doses of radiation.[11,12,17,18]

A matched case–control study comparing surgical resection with cobalt plaque brachytherapy has been published,[13] but surgical techniques have changed since then, and cobalt has been replaced with other radioisotopes. A nonmatched comparison of TSR and IBT has also appeared.[17,19] Both reports concentrated on survival and visual outcome, and as single-center studies they may have been subject to selection bias in choosing 1 of the 2 treatments for an individual patient.[13,17] In the present study, treatment was assigned based on the center where the patient was seen, rather than tumor characteristics, and the full spectrum of treatment-related complications was compared in addition to tumor control and visual outcome.

Patients and Methods

Study Aims

The primary aim was to compare local tumor control, complications, and visual loss after IBT and TSR by matched case–control design among patients who might be treated with either method. Secondary aims were to assess survival rate and quality of life.

Eligibility Criteria

Patients were eligible for this study if they had a primary choroidal melanoma that was ≥6 mm thick. Iodine brachytherapy was performed at the Department of Ophthalmology, Helsinki University Central Hospital, Helsinki, Finland (1990–1998), and TSR was done at the Tennent Institute of Ophthalmology, Glasgow (1988–1992) and St. Paul's Eye Unit, Royal Liverpool University Hospital, Liverpool (1993–1998), United Kingdom.

During the study period, choroidal melanomas <6 mm thick were treated as a rule by RuBT in each center. Primary enucleations were performed based on factors such as age and general health of the patient, tumor size and location, vision in both eyes, retinal detachment, and patient preference. Choroidal melanomas ≥6 mm thick that were not enucleated were treated as a rule by IBT in Helsinki and, if the patients were fit for hypotensive anesthesia, by TSR in Glasgow and Liverpool.

Matching Variables

The patients were matched in November 1998 according to 4 baseline variables, considered to have a major impact on chances of local recurrence, complications, and visual loss:

1. Age at diagnosis: (1) <55 years versus (2) ≥55 years.
2. Best-corrected Snellen acuity: (1) ≥20/40 versus (2) 20/50 to 20/200 versus (3) <20/200. Visual acuity was consistently tested with best spectacle correction from a distance of 6 m, but the test types varied between the 2 centers and over time.
3. Clinical estimate of largest basal tumor diameter (LBD) derived from binocular indirect ophthalmoscopy and B-scan ultrasonography: (1) <12 mm versus (2) 12 to 14 mm versus (3) >14 mm.
4. Distance between the posterior tumor margin and the optic disc or foveola: (1) ≤2 disc diameters (i.e., approximately 3.0 mm) from either structure versus (2) >2 disc diameters from both, evaluated from fundus photographs.

Efficient matching variables should be strongly associated with outcome but unassociated with each other.[20] Because LBD and tumor height are related, height was not matched, expecting that matching for LBD would sufficiently reduce bias due to tumor height.

A 4-digit index for each patient was computer generated from the matching variables, and patients were then matched on the basis of the index, masked to other pretreatment and outcome data. Patients were matched only if treated within 3 years of each other so as to balance follow-up times, but the length of follow-up itself was not a matching criterion. If more than 1 patient matched an index case, the one treated closest to the index case was selected.

For 49 of 80 eligible patients in the IBT group a match was found from the 166 eligible patients in the TSR group; these figures included all patients who had choroidal melanoma ≥6 mm thick managed by either method in the corresponding center during the study period. The main reason for not finding a match was the large size of remaining melanomas managed by IBT.[21] The year of treatment was comparable for the matched pairs (median = 1994; $P = 0.11$, Wilcoxon signed rank test).

Outcome Variables

Time to and reason for the following outcomes had prospectively been assessed and stored in computerized databases in both participating centers. Each principal investigator visited the other center to compare methods of clinical evaluation and database structure before the data were combined, so as to maximize comparability in recording outcomes.

1. Local recurrence, categorized by type[22] and location relative to the tumor. An unequivocal 1.0 mm increase in lateral tumor dimension or tumor thickness or appearance of new tumor in a previously uninvolved area was taken to indicate recurrence.
2. Death, coded as being due to melanoma, other cancer, or nonneoplastic disease. Both participating centers routinely have metastases confirmed by histopathology, if appropriate, and have legal access to their corresponding population data registry and death certificates to assess life outcome.
3. Clinically significant lens opacity, evaluated by slit-lamp biomicroscopy and defined as a new lens opacity, typically posterior subcapsular cataract in type, which decreased or would have decreased visual acuity if vision was reduced because of other reasons.

Table 1. Comparison of Baseline Characteristics of 49 Matched Pairs of Patients Who Underwent Transscleral Local Resection and Iodine Brachytherapy for Malignant Uveal Melanoma

Characteristic	Transscleral Local Resection	Iodine 125 Plaque Brachytherapy	*P*
Matching variables			
Median age [yrs, (range; interquartile range)]	55 (25–73; 47–61)	57 (31–87; 46–67)	0.0016*
Median LBD [mm (range; interquartile range)]	14.0 (7.0–20.0; 11.0–15.0)	13.3 (6.5–22.0; 11.0–15.0)	0.22*
Median visual acuity (range; interquartile range)	20/40 (20/16–HM; 20/30–20/100)	20/40 (20/16–HM; 20/30–20/100)	0.27*
Other baseline variables			
Male:female	23:26	32:17	0.10†
Tumor height [N (%)]			
≤8 mm	29 (59)	30 (61)	0.83†
>8 mm	20 (41)	19 (39)	
Median tumor height [mm (range; interquartile range)]	8.0 (6.0–13.0; 7.0–9.0)	7.8 (6.0–14.0; 7.0–9.3)	0.75*
COMS classification [N (%)]			0.41†
Medium	30 (61)	35 (71)	
Large, by height	9 (18)	8 (16)	
Large, by LBD	7 (14)	3 (6)	
Large, peripapillary	3 (6)	3 (6)	
Cell type [N (%)]			NA
Spindle	25 (51)	NA	
Mixed and epithelioid	24 (49)	NA	

COMS = Collaborative Ocular Melanoma Study; HM, hand movements; LBD = largest basal tumor diameter; NA = not applicable.
*Wilcoxon signed rank test, 2-sided.
†Stuart–Maxwell test, 2-sided.

4. Neovascularization of the iris or chamber angle, evaluated by slit-lamp biomicroscopy and, if suspicious vessels were seen or intraocular pressure was elevated, by gonioscopy.
5. Persistent increase in intraocular pressure over 24 mmHg by applanation tonometry.
6. Vitreous hemorrhage, evaluated by binocular indirect ophthalmoscopy.
7. Retinal detachment, categorized as rhegmatogenous or exudative, evaluated by binocular indirect ophthalmoscopy.
8. Maculopathy, categorized as scarring, macular pucker, radiation maculopathy, or cystic macular edema, evaluated by slit-lamp biomicroscopy using a contact or noncontact lens.
9. Optic neuropathy, categorized as radiation optic neuropathy and optic atrophy; optic neuropathy was evidenced by edema, peripapillary splinter hemorrhages, and microinfarcts.
10. Loss of visual acuity levels of (1) 20/60 (corresponding to loss of reading vision from the study eye) and (2) 20/200 (corresponding to legal blindness of the study eye).
11. Quality of life was provisionally assessed by a structured 14-item interview on visual function and patient satisfaction (Table 3). This assessment was conducted in 1999 after the patients had been matched, at which time the questionnaire was answered by 11 of the matched pairs (inclusion ratio = 22%) and by 18 additional patients whose matched case subject had died or could not be reached (inclusion ratio = 41%).

Treatment and Follow-up

Iodine brachytherapy was administered using custom-made, nonrimmed gold plaques (diameter = 15, 20, and 25 mm; thickness = 0.5 mm).[23] For tumors close to the optic nerve, a notched plaque was used (diameter = 20 mm). The tumor margins were identified under local or general anesthesia using transscleral illumination and indirect ophthalmoscopy. A minimum safety margin of 2 mm was a goal but not an absolute requirement. Tumors extending close to the foveola were often managed with an eccentric plaque with a smaller posterior safety margin. Of the brachytherapies, 24 were performed by a single surgeon in the ocular oncology service and 25 by any of 5 surgeons in the vitreoretinal service.

The prescription dose was usually 100 Gy to the apex of the tumor, allowing 1 mm for the thickness of the sclera (this dose is currently 80 Gy, prescribed to 9 patients in this series). A median dose of 98 Gy (range = 55–126) was delivered within a median of 119 hours (range = 48–360). The median dosage rate was 77 cGy/hour (range = 17–182).

Transscleral resection was performed according to previously described techniques.[11,24] Hemorrhage was minimized by systemic hypotensive anesthesia. A partial-thickness scleral flap was prepared over the tumor, which was resected together with deep scleral lamella and a surround of normal choroid, if possible, without damaging retina. All procedures were performed by the same surgeon. Adjunctive RuBT (20-mm plaque, dose = 500 Gy to the sclera or 100 Gy to 2–3 mm) was administered in 16 patients (including 14 of 18 enrolled patients treated in 1995–1998).[25] Ocular decompression by partial 1-port vitrectomy was performed in more recent operations.[24]

In both participating centers, patients were reviewed approximately 1 month after therapy, then at 6 monthly intervals for 3 to 6 years, and annually thereafter. Because of differences in health care systems, the follow-up times of patients treated with TSR were shorter than those of patients treated with IBT (median = 4.6 vs. 5.6 years; $P<0.001$, Wilcoxon signed rank test). We evaluated this difference as a potential cause of bias and found no evidence that the length of follow-up would have depended on the risk of developing the outcomes of interest. Because censoring appeared random and survival analysis then provides an unbiased estimator of survival,[26] the difference in follow-up should not appreciably bias the analysis presented. Indeed, reversing the situation by deleting twice the difference in follow-up from patients treated with IBT did not change the position of the survival curves.

At each visit, a full ophthalmic examination was performed,

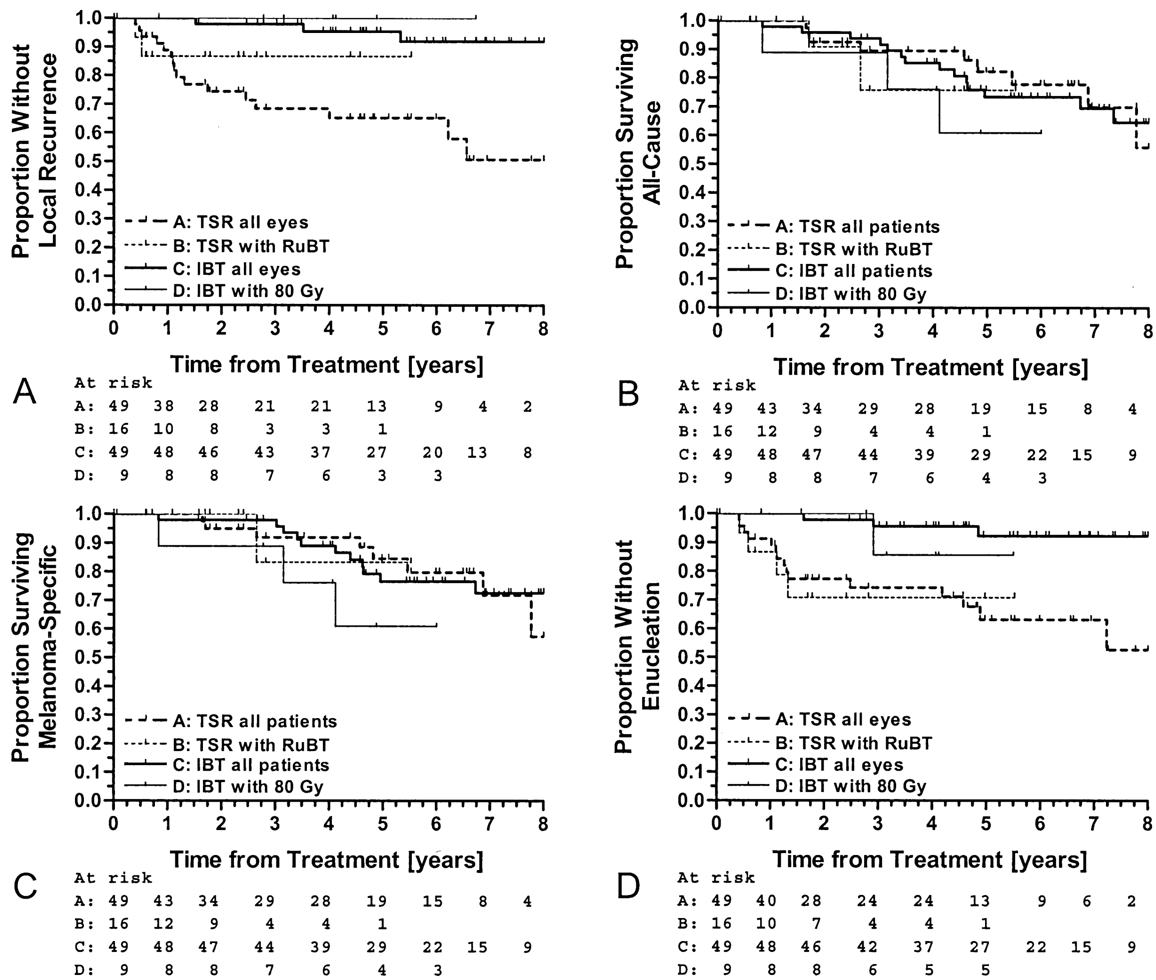

Figure 1. Survival and local tumor control for 49 matched pairs whose choroidal melanoma was treated with transscleral local resection (TSR) and iodine brachytherapy (IBT). The Kaplan–Meier plots show observed local tumor recurrence (**A**), all-cause mortality (**B**), melanoma-specific mortality (**C**), and secondary enucleation (**D**). The numbers below the graphs show patients at risk. Note comparable survival in face of a difference in local recurrence. The subgroups that received adjuvant ruthenium brachytherapy (RuBT) at the time of TSR and that were treated with 80 Gy of IBT are shown for comparison.

including measurement of tumor thickness by B-scan ultrasonography and monitoring of lateral tumor margins by comparing the ophthalmoscopic appearance to previous photographs. Thickness was measured in both centers from the top of the tumor to the inner surface of the sclera. Follow-up data were collected until June 1, 2001.

Statistical Analysis

The data were analyzed with the Stata (Release 7.0, Stata Co., College Station, TX) and StatXact-5 (Cytel Software, Cambridge, MA) statistical packages. Descriptive statistics are given as the median and range. Continuous variables were compared using the Wilcoxon signed rank test for matched pairs, and categorical variables were compared using the Stuart–Maxwell chi-square test for matched pairs.[27] Nonpaired categorical data were compared with the Kruskal–Wallis test.

Kaplan–Meier survival curves were visually compared.[28] Data for patients treated with TSR and adjunctive RuBT[25] and IBT with a prescription dose of 80 Gy were plotted separately for descriptive purposes. The present study was not designed to formally compare outcome with and without adjunctive RuBT or according to the dose of IBT.

Because routine analysis of time-to-event data does not take into account correlation between subjects, which is inherent in matched case–control study design, we used extensions of parametric regression, which allow for dependence between events.[26,29] Statisticians do not currently agree on the best method to analyze correlated survival data. We used 2 methods of analysis recommended for this purpose.[26,29] Because of limited sample size and the exploratory nature of the study, variables with *P* values of <0.10 should be considered for future analysis.

We fitted a marginal Weibull parametric regression model,[28,29] which accounted for dependence between the members of matched pairs by clustering on the pair.[26,29] Each member was at risk for failure both before and after his or her companion had failed, and the risk set consisted of all patients who had not yet failed (i.e., the analysis was based on unordered events of the same type).[26] The marginal model estimates the averaged relative risk due to treatment.[26]

As a secondary analysis, we fitted a generalization of the Weibull model, which can account for presence of a latent, unobservable, multiplicative effect on the hazard function.[26,28,29] This is known as a frailty model, because it considers some individuals to be more "frail" than others (i.e., they have a higher chance of failing for reasons not captured by the known variables that can be

Table 2. Weibull Parametric Regression of Time to Adverse Event Among 49 Matched Pairs of Patients Who Underwent Transscleral Local Resection (TSR) or Iodine 125 Brachytherapy (IBT) for Malignant Uveal Melanoma

	Marginal Model			Shared Frailty Model				
Dependent Variable	*Hazard Ratio (SE)*	*95% CI*	P	*Hazard Ratio (SE)*	*95% CI*	P	θ	P
Local recurrence	0.12 (0.07)	0.04–0.36	<0.001	0.02 (0.02)	0.01–0.11	<0.001	5.36	0.005
All-cause mortality	1.10 (0.44)	0.51–2.41	0.80	1.07 (0.49)	0.43–2.63	0.89	1.83	0.034
Melanoma mortality	0.97 (0.42)	0.42–2.24	0.94	0.81 (0.39)	0.30–2.22	0.69	2.66	0.023
Cataract	2.05 (0.67)	1.08–3.89	0.029	2.04 (0.58)	1.18–3.55	0.011	<0.001	1.00
Vitreous hemorrhage	2.30 (1.04)	0.95–5.57	0.064	2.30 (1.01)	0.97–5.45	0.058	<0.001	1.00
Retinal detachment	0.84 (0.31)	0.40–1.75	0.63	0.83 (0.32)	0.39–1.76	0.61	0.67	0.12
Maculopathy	2.28 (1.01)	0.96–5.43	0.062	2.28 (0.89)	1.07–4.89	0.034	<0.001	1.00
20/60 vision	1.37 (0.41)	0.76–2.45	0.29	1.32 (0.43)	0.70–2.50	0.39	0.12	0.35
20/200 vision	2.38 (0.58)	1.48–3.83	<0.001	2.38 (0.64)	1.41–4.04	0.001	<0.001	1.00

CI = confidence interval; HR = hazard ratio; θ = frailty variance, used to estimate presence of significant frailty effect—i.e., correlation within matched pairs (present if $P < 0.05$); SE = Standard error.
HR is adjusted for matching either by clustering on matched pair (the marginal model, estimates the averaged relative risk due to treatment) or by shared frailty (the shared frailty model, estimates the relative risk due to treatment within matched pairs). Coding: TSR = 0, IBT = 1 (HR<1 indicates higher risk for TSR, HR>1 indicates higher risk for IBT; significant difference if $P<0.10$).

entered into the model).[30] Sharing a frailty value generates dependence between individuals, which can be used to model intragroup correlation.[29,30] Because one can reasonably expect that the time to event of the members of a given matched pair are correlated, we analyzed frailty as being shared on the matched pair level. We used the γ frailty distribution, for which the relative variability of frailty values for survivors is constant over time.[26,28,30] Using the inverse Gaussian distribution, which makes frailties of survivors more homogenous with passage of time,[30] produced comparable results (data not shown). The frailty model estimates the relative risk due to treatment within matched pairs,[26] corresponding to the study design. In addition to hazard ratios (HRs), this model estimates frailty variance, θ, which can be used to test for the presence of a significant frailty effect.[30] When within-pair correlation is zero, the HR estimated by the frailty model will be identical to that of the marginal model (i.e., the frailty model does not provide advantage over a model that does not take frailty into account). When θ is significant, evidence of frailty (in this case, evidence that the risk of failing was correlated within matched pairs) is present, and the frailty model should be preferred to get a better estimate of the hazard due to treatment (TSR vs. IBT).

The sample of this retrospective study to determine effect size was determined by the maximum number of pairs that could be matched. Projected sample sizes for a potential prospective randomized study were calculated with the -calcssi- module available for Stata.[31]

Results

The TSR and IBT groups were comparable with respect to gender (Table 1). Of the matched pairs, 27 (55%) were 55 years or older. The median age was comparable for the 2 groups (Table 1), but the age distribution of patients treated with IBT was skewed toward higher ages ($P = 0.0016$, Wilcoxon signed rank test).

The LBD was less than 12 mm for 13 (26%) matched pairs, between 12 and 14 mm for 17 pairs (35%), and 14 mm or more for 19 pairs (39%). The groups were comparable with regard to tumor height and Collaborative Ocular Melanoma Study category[9,16] (Table 1). The median difference in tumor height within the matched pairs was 0 mm (interquartile range = −1.0 to +1.4 mm). The posterior tumor margin was within 2 disc diameters of the optic disc, macula, or both structures for 30 (61%) pairs. The visual acuity was 20/40 or better for 17 (35%) matched pairs, between 20/50 and 20/200 for 25 pairs (51%), and worse than 20/200 for 7 pairs (14%).

Local Tumor Control and Survival

Three patients developed a local tumor recurrence after IBT (2 vertical, 1 marginal), and 16 patients developed a recurrence after TSR (8 at the rim of and 4 within the resection area, 2 elsewhere in the eye, and 2 in more than one of these locations; 2 patients also developed an additional extrascleral recurrence). Of the 16 patients with local recurrence after TSR, 14 had not received adjuvant RuBT (Fig 1A). The risk of local recurrence was significantly smaller after IBT (Fig 1A, Table 2; HR = 0.12, $P<0.001$, Weibull regression, marginal model, averaged HR for paired data). The frailty model revealed significant correlation within pairs (θ = 5.36, $P<0.001$; Table 2) and suggested that the risk difference, estimated within matched pairs, was greater (HR = 0.02; Table 2).

Nine patients in the TSR group and 11 patients (all histopathologically confirmed) in the IBT group died of metastatic uveal melanoma. One patient in the TSR and 2 patients in the IBT group died of nonneoplastic disease (including one autopsy-proven cerebrovascular hemorrhage), and one patient died of autopsy-confirmed astrocytoma. All-cause and melanoma-specific survival after TSR and IBT were comparable (Fig 1B, C; HR = 1.10 and 0.97, respectively; $P>0.10$; Table 2). The frailty model indicated significant correlation within pairs (θ = 1.83 and 2.66, respectively; $P<0.05$) but led to similar conclusions with regard to effect size (HR = 1.07 and 0.81, $P>0.10$; Table 2). Of 16 patients who developed local recurrence after TSR, 3 died of metastases within 2 years of recurrence and 9 others had been observed for 3 to 5 years after recurrence without evidence of metastasis. The remaining patients had a shorter follow-up than 3 years after recurrence. The patients who had local recurrence after IBT were alive and without metastases at 2, 4, and 5 years after recurrence.

Indications for secondary enucleation (Fig 1D) differed between the 2 centers. Two eyes were removed after IBT because of complications, and one because of recurrence. After TSR, local tumor recurrence and untreatable retinal detachment were routinely managed with enucleation.

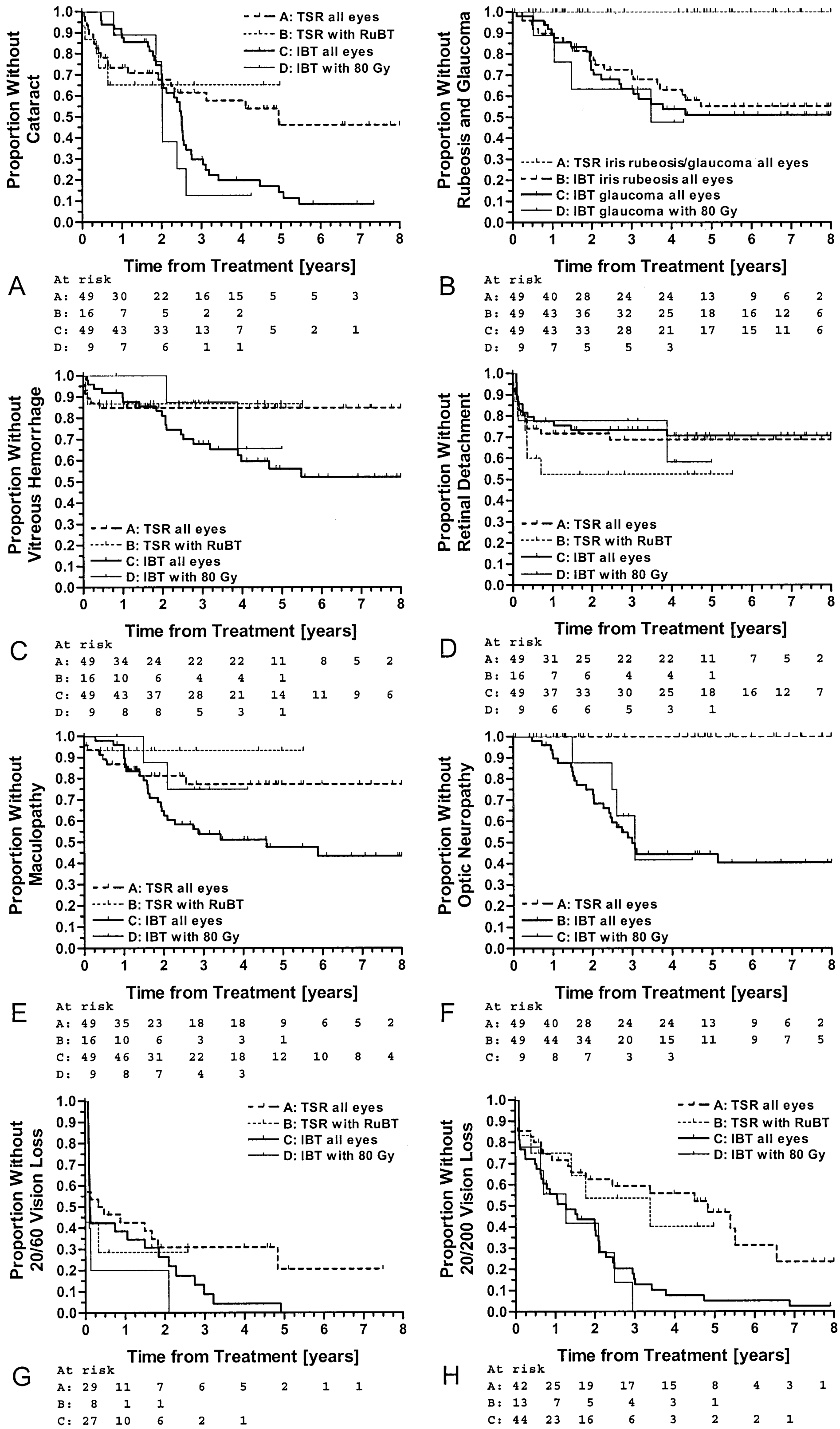

Figure 2. Complications and visual outcome for 49 matched pairs whose choroidal melanoma was treated with transscleral local resection (TSR) or iodine brachytherapy (IBT). The Kaplan–Meier plots show observed time to development of cataract (**A**), rubeosis and glaucoma (**B**), vitreous hemorrhage (**C**), retinal detachment (**D**), maculopathy (**E**), optic neuropathy (**F**), and loss of 20/60 (**G**) and 20/200 vision (**H**). The numbers below the graphs indicate patients at risk. Note that complications were more common after iodine brachytherapy. The subgroups that received adjuvant ruthenium brachytherapy (RuBT) at the time of TSR and that were treated with 80 Gy of IBT are shown for comparison.

Table 3. Quality-of-Life Questionnaire Outcome among 16 Patients Who Underwent Transscleral Local Resection and 24 Patients Who Underwent Iodine Brachytherapy for Malignant Uveal Melanoma, Including 11 Matched Pairs

Item	Transscleral Local Resection				Iodine 125 Plaque Brachytherapy				P*
	Not at All	*A Little*	*Quite a Bit*	*Very Much*	*Not at All*	*A Little*	*Quite a Bit*	*Very Much*	
1. Pain, soreness, grittiness, or discomfort	10 (63)	6 (37)	0 (0)	0 (0)	11 (44)	12 (48)	2 (8)	0 (0)	0.19/0.66
2. Dissatisfaction with cosmetic result	16 (100)	0 (0)	0 (0)	0 (0)	17 (74)	4 (17)	1 (4)	1 (4)	0.03/0.08
3. Troubled by defects in side vision	8 (50)	5 (31)	2 (13)	1 (6)	10 (42)	6 (25)	2 (8)	6 (25)	0.35/0.46
4. Troubled by double vision	10 (63)	5 (31)	1 (6)	0 (0)	19 (80)	4 (17)	0 (0)	1 (4)	0.27/0.17
5. Troubled by distorted vision	5 (36)	6 (43)	2 (14)	1 (7)	13 (60)	5 (23)	3 (14)	1 (5)	0.26/0.95
6. Interference with vision of the fellow eye	12 (75)	4 (25)	0 (0)	0 (0)	7 (28)	16 (64)	1 (4)	1 (4)	<0.01/0.03
7. Difficulty driving because of poor vision	9 (69)	4 (31)	0 (0)	0 (0)	7 (44)	9 (56)	0 (0)	0 (0)	0.08/0.90
8. Difficulty seeing television	13 (81)	2 (13)	1 (6)	0 (0)	15 (60)	8 (32)	0 (0)	2 (8)	0.17/0.41
9. Difficulty walking in crowded areas	9 (56)	4 (25)	1 (6)	2 (13)	7 (29)	15 (60)	1 (4)	2 (8)	0.24/0.08
10. Difficulty reading because of poor vision	10 (58)	5 (35)	0 (0)	1 (6)	8 (32)	13 (52)	2 (8)	2 (8)	0.06/0.41
11. Worry about intraocular recurrence	6 (37)	7 (44)	0 (0)	3 (19)	5 (20)	16 (64)	0 (0)	4 (16)	0.46/0.72
12. Worry about dying because of the tumor	7 (47)	4 (27)	3 (20)	1 (7)	11 (44)	9 (36)	3 (12)	2 (8)	0.96/0.32
13. Worry about losing the eye	7 (44)	5 (31)	0 (0)	5 (25)	9 (36)	9 (36)	6 (24)	1 (4)	0.93/0.17
14. Regret about trying to save the eye	15 (94)	0 (0)	0 (0)	1 (6)	24 (96)	1 (4)	0 (0)	0 (0)	0.72/1.00

*Kruskal–Wallis nonpaired test based on all 40 patients/Stuart–Maxwell paired test based on 11 matched pairs (22 patients).

Treatment-Related Complications

The risk of developing cataract was significantly smaller after TSR than after IBT (Fig 2A; HR = 2.05, P = 0.029; Table 2). Main reasons for cataract were radiation damage after IBT and vitrectomy, often with silicone oil, after TSR. The frailty model did not suggest significant dependence within pairs with regard to time to this or any other complication (Table 2).

Rubeosis and glaucoma (Fig 2B) developed only after IBT. Of 22 patients with treatment-related glaucoma, 16 developed neovascular glaucoma and 6 developed another type of secondary glaucoma (phakomorphic glaucoma, secondary congestive and open-angle glaucoma). Even though several eyes with neovascular glaucoma lost light perception, few were enucleated. This was because of the policy of the center to retain eyes for cosmetic reasons with medical treatment and transscleral laser cyclophotocoagulation.[32]

The risk of vitreous hemorrhage was higher after IBT (Fig 2C; HR = 2.3, P = 0.064). This complication occurred only within 6 months from TSR, whereas it could develop several years after IBT, presumably from the degenerating tumor. Because of grossly different risk over time, the borderline HR and frailty variance (P = 0.12; Table 2) likely reflect poor fit. The risk of retinal detachment (Fig 2D; HR = 0.84, P = 0.63) did not differ between TSR and IBT. All 14 detachments after IBT were exudative, and most were transient. Of 14 detachments after TSR, 2 were exudative and 12 rhegmatogenous; 5 of the latter were successfully reattached.

Maculopathy (Fig 2E; HR = 2.28, P = 0.062; Table 2) was more frequent after IBT. This complication developed in 25 patients after IBT (17 had radiation retinopathy, 3 had lipid exudation from leaky tumor vessels, 4 had macular scarring from irradiation, and 1 developed macular pucker) and in 9 patients after TSR (4 were due to surgery, 4 to macular pucker, and 1 to cystic macular edema). One patient developed evidence of radiation retinopathy after TSR and adjuvant RuBT without loss of vision.

Optic neuropathy (Fig 2F) occurred only after IBT. Of 25 patients who developed this complication, 16 showed active radiation optic neuropathy with bleeding and microinfarcts, and 9 directly developed secondary optic atrophy. In addition, 4 developed neovascular glaucoma. Exploratory subgroup analysis suggested that neuropathy was not less frequent if the tumor was located outside 2 disc diameters from the optic disc than within this distance (data not shown).

Visual Outcome and Quality of Life

The risk of losing 20/60 vision did not differ statistically after TSR and IBT (Fig 2G; HR = 1.37, P = 0.29; Table 2), but no more than 31 pairs had this acuity level before treatment. The risk of losing 20/200 vision was significantly higher after IBT than after TSR (Fig. 2H; HR = 2.38, $P < 0.001$). Exploratory subgroup analysis suggested a smaller risk of losing 20/60 vision after TSR if the tumor was located more than 2 disc diameters from the optic disc and foveola (data not shown).

The 14-item quality-of-life questionnaire suggested more problems related to appearance, interference from the treated eye, and walking in crowded areas after IBT (items 2, 6, and 9), but it gave no indication of a major overall difference between the 2 treatments (Table 3).

Discussion

Iodine brachytherapy of medium-sized and large uveal melanomas leads to many complications, which reduce the possibility of saving useful vision.[8,21] Transscleral resection is one alternative to IBT that has been developed in an attempt to achieve better visual results without compromising local tumor control and survival.[11,12] Especially with the development of methods for preventing 2 main complications of TSR—local tumor recurrence[18,25] and rhegmatogenous retinal detachment[33]—there would now be scope to compare these 2 methods prospectively. The present study provides estimates of complication rates and effect sizes so that randomized, controlled studies can be considered.

A matched case–control study is helpful in reducing bias, but it has important limitations. Patients selected by the matching process may not represent the entire underlying patient population. It is possible that some patients who fulfill the inclusion criteria of the present analysis might fare better with one treatment than the aggregated results seem to indicate. Matching according to 2 to 3 categories of a variable, as was done here, is estimated to remove roughly 60% to 80% of bias due to that variable, respectively, but matching neither eliminates bias completely nor adjusts for unknown confounders.[20] The latter can only be controlled by randomization. Matching according to baseline characteristics does not control for differences in follow-up. In this study, follow-up routines were similar, but management differed once complications developed, especially as regards secondary enucleation. Consequently, we do not aim to demonstrate the superiority of one treatment over the other. The goal is to provide an estimate of what might be expected in a subgroup of patients who can be treated by either technique.

In our data set, the risk of local recurrence was 8 times lower after IBT than after TRS. The cumulative rate of recurrence at 5 years after IBT was roughly similar to previous reports.[34–36] The relatively high risk after TSR is a well-recognized problem,[18] especially with posterior tumor extension, which was overrepresented in this matched series, so that resection is now often routinely combined with adjuvant RuBT.[17,25] Noncontrolled data suggesting that the recurrence rate after adjuvant RuBT is not notably higher than that after IBT still need to be confirmed.[17,18]

Despite the higher risk of local recurrence after TSR, no difference was observed in melanoma-specific 8-year survival, unlike that which would be expected on the basis of studies that associate recurrence after radiotherapy with higher mortality.[22,37,38] It may be that melanomas that recurred after TSR were more aggressive. Thus, the matched pair treated with IBT also had micrometastases, and both patients thus died irrespective of treatment. Alternatively, a generally small, marginal recurrence due to surgical miss[22] after TSR did not affect life outcome when managed with prompt enucleation.

Several complications such as cataract, vitreous hemorrhage, retinal detachment, and maculopathy occurred after either treatment, but after different intervals. Most patients treated with IBT eventually developed cataract, which usually appeared from 1 to 3 years after treatment, on average later than after TSR. Vitreous hemorrhage occurred only in the first few months after TSR. The relative risk was strongly time dependent, and the regression model did not efficiently capture it, underscoring the importance of visually comparing observed survival when interpreting results of regression analyses.

Other complications seen after IBT were essentially nonexistent in matched patients treated with TSR. Glaucoma and rubeosis[17] never occurred after TSR in this series because all tumors were choroidal and the resection thus did not involve the angle and ciliary body, and because of a policy of enucleating when untreatable retinal detachment developed. Almost half of the patients treated with IBT developed glaucoma,[17] predominantly neovascular in type and presumably caused by radiation damage and factors related to the presence of a sizable dying tumor in the eye. Radiation maculopathy and neuropathy after IBT were common regardless of tumor distance from the optic disc and foveola.[39] These complications would not be expected to occur after adjuvant RuBT, unless the tumor is within 2 disc diameters of these structures.[40] The relatively high frequency of complications is explained in part by the fact that the study concerns a selected subset of patients, many of whom had high-risk characteristics such as tumors classified as large in the Collaborative Ocular Melanoma Study protocol[41] and tumors that were located close to the optic disc, a fact that also made it worthwhile to seek alternatives to radiotherapy.

The risk of losing 20/200 vision was 2.4 times greater after IBT. Based on the corresponding Kaplan–Meier plot, the estimated effect size was 0.44 at 3 years (0.59 for TSR vs. 0.15 for IBT), which is essentially identical to the difference of 0.43 (0.58 vs. 0.15, respectively) estimated from a nonmatched series of choroidal and ciliary body melanoma.[17] This pilot study did not have adequate power to detect small differences in retaining 20/60 vision. Exploratory subgroup analysis suggested, however, that TSR might be more effective than IBT in conserving 20/60 vision if the tumor does not extend to within 2 disc diameters of the disc or foveola. In fact, the present matched analysis may not reflect the full potential of conserving 20/60 vision by TSR because of a high number of tumors extending close to the disc and foveola.

Local tumor control after TSR is improving with adjuvant RuBT,[17,25] and retinal detachment may also become less problematic with developments in vitreoretinal surgery.[33] Transscleral resection remains a demanding technique associated with risks of both immediate visual loss and complications of hypotensive anesthesia.[11] Plaque brachytherapy is technically easier and can be performed under peribulbar anesthesia, being possible even if the patient cannot be considered for TSR, but it is less likely that complication rates will notably decrease in the future. In theory, the chances of radiation maculopathy and neuropathy are diminished with collimated plaques,[42,43] optimized seed positioning,[44] isotope selection,[10] and dosage reduction, and the risk of neovascular glaucoma might additionally be prevented by treating leaking tumors with transpupillary thermotherapy or resection. In practice, little advantage has been reported from collimated plaques,[17] and dosage reduction increases risk of local recurrence.[34]

Only a randomized, prospective trial can prove that either of the 2 treatments is better than the other. If such a study of TRS and IBT for choroidal melanoma 6 mm or thicker were to become feasible, outcome measures would likely be local tumor control as a surrogate of melanoma-related death, and retention of 20/60 vision. The present study suggests a clinically meaningful difference of approximately 0.25 at 3 years for loss of 20/60 vision (0.31 vs. 0.09; possibly greater if tumors within 2 disc diameters from disc and foveola were ineligible). It can be estimated that 175 patients per arm would be required to confirm this difference with a power of 0.80 (HR = 1.4 from Table 3; recruitment = 3 years; study duration = 6 years).[31] This would also be sufficient to confirm the 0.28 difference in local recurrence at 3 years, suggested by the Kaplan–Meier plot (0.32 vs. 0.04; HR = 0.12, power = 0.99), whereas 315 patients per arm would be needed if adjuvant RuBT were to reduce the difference to 0.10. Studies of this size can be conducted only as multicenter trials. Based on our results, it also remains to be confirmed whether local recurrence after TSR is biologically equivalent to local recurrence after IBT and, consequently, a legitimate surrogate for melanoma-related mortality, because such a regrowth after a surgical miss is likely to occur irrespective of the aggressiveness of the primary tumor.

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